

Repair Request - Requirement & Scope

Business Baseline + Performance NFR Clarification

Document Control	Value
Document ID	RR-REQ-001
Version	1.6 - Performance Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Supersedes	RR-REQ-001 v1.5
Revision Scope	Clarify measurable performance boundary and cross-document traceability. No new workflow/state/role/SLA semantics.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

ระบบแจ้งซ่อม / REPAIR REQUEST

Requirement & Scope Document / Baseline Revision

Document Control	Value
Document ID	RR-REQ-001
Version	1.5 - Baseline Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	Prior Requirement v1.4 + Audit closure decisions D-11..D-15
Audience	Business Owner, SA, Developer, Tester, Project Manager
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด business scope, requirements, locked decisions, governance and traceability for Repair Request MVP.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. เป้าหมายธุรกิจ / Business Outcome

ระบบ Repair Request ครอบคลุมวงจรตั้งแต่สร้างคำขอ ตรวจสอบ/อนุมัติ สร้าง Work Order จัด Service Visit ปฏิบัติงาน ตรวจสอบรับ Corrective Action ปิดงาน ติดตาม Resolution SLA, audit และรายงาน เพื่อให้ทุกธุรกรรมมี owner, state, evidence และประวัติที่ตรวจสอบย้อนหลังได้.

2. ขอบเขต / Scope Boundary

In Scope	Out of Scope
• Repair Request + attachment • Review / Approve / Reject / Return • Work Order / Service Visit / Schedule / assignment • Work Session: Check-in/Pause/Resume/Check-out • Work Summary / Cost Summary / Customer Acceptance / Corrective Action • Resolution SLA, Risk/Breach, Override, Time Correction • Audit, in-system notification, MVP reports/export	• Spare-part purchasing/accounting • GPS route optimization • Vendor payment/procurement • Full offline synchronization • AI diagnosis/predictive maintenance • External integration and Response SLA

3. ผู้ใช้งานและความรับผิดชอบ / Users and Responsibilities

Role	Responsibility / Permission Boundary
Requester	สร้าง/แก้ไข Draft, Submit, ดูข้อมูลของตน, Cancel Repair Request ก่อน Convert
Approver	Review / Approve / Reject / Return for Correction ตาม route และ Site/หน่วยงาน
Coordinator	Convert to Work Order, Schedule/Assign/Reassign/Reschedule/Cancel Visit, decide MISSED
Technician	Work Session ของ Visit ที่ได้รับมอบหมาย: Check-in/Pause/Resume/Check-out, evidence
Team Lead	Submit Work Summary, Cost Summary, request Time Correction, Corrective Plan
Supervisor	Review summary/cost, approve/reject Time Correction, approve Corrective Plan, cancel/close WO, authorized SLA Override
Customer Acceptance Contact	Requester หรือ Site Contact ที่ถูก designate เพียงคนเดียว; Accept/Reject work
Administrator	Master data / role / site scope / SLA configuration; ไม่ impersonate business actor โดยอัตโนมัติ

4. Functional Requirements

ID	Name	Requirement
FR-01	Create/Edit Repair Request	Requester สร้างและแก้ไข Draft พร้อม Site, Equipment/Location, Category, Priority, Problem, Contact, Preferred Window และ Attachment.
FR-02	Validate & Submit	Server ตรวจสอบ mandatory/format/active master/scope/duplicate ก่อน Submit; duplicate warning ทำต่อได้เมื่อมี continuation reason.
FR-03	Approval	Route, Approve, Reject, Return for Correction พร้อม decision history, reason, audit, notification.
FR-04	Convert to Work Order	

ID	Name	Requirement
		Coordinator สร้าง Work Order ได้หนึ่งรายการจาก APPROVED Request; transaction atomic; Request -> CONVERTED.
FR-05	Service Visit	Schedule, Assign/Reassign, Reschedule, Cancel, MISSED และ follow-up ตาม state/role.
FR-06	Work Session	Assigned Technician Check-in/Pause/Resume/Check-out; no overlap; summary/outcome/evidence guards.
FR-07	Acceptance & Corrective	Designated contact Accept/Reject; Reject creates Corrective Action and re-acceptance cycle.
FR-08	SLA & Time	Resolution SLA 24h 24/7, Risk at T-4h, breach, Override, Time Correction; correction does not recalc SLA.
FR-09	Search/Audit/Reports	Authorized users search/list/view timeline/audit/report/export within tenant/site/role scope.

5. Workflow & State Baseline

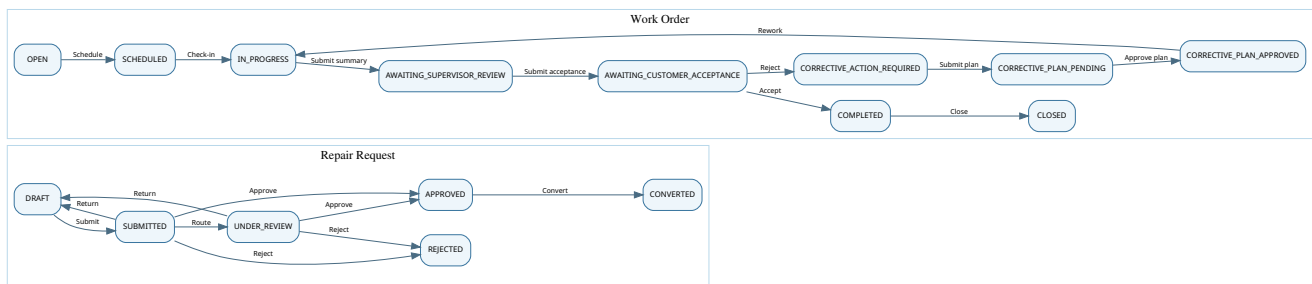


Figure 1 - State overview. Cancel paths and detailed guard conditions are defined in RR-STS-001 v1.5.

Repair Request: DRAFT → SUBMITTED → UNDER_REVIEW (optional) → APPROVED → CONVERTED. Return for Correction is an **action** to DRAFT; no RETURNED state. REJECTED/CANCELLED are terminal.

Work Order: OPEN → SCHEDULED → IN_PROGRESS → AWAITING_SUPERVISOR_REVIEW → AWAITING_CUSTOMER_ACCEPTANCE → COMPLETED → CLOSED; Customer Reject opens corrective states and loops back through review/acceptance. CLOSED/CANCELLED have no Reopen in MVP.

Service Visit: SCHEDULED / RESCHEDULED / IN_PROGRESS / COMPLETED / MISSED / CANCELLED. MISSED original record remains MISSED per D-15.

6. Core Business Rules

ID	Rule
BR-01	Request No. unique; create/audit identity immutable.
BR-02	Backend validates mandatory, format, master, role, tenant/site, ownership, assignment and current state.
BR-03	Create Work Order only from APPROVED; exactly one Work Order per Repair Request; Request becomes CONVERTED.
BR-04	Reject, Cancel, Reassign, Reschedule, SLA Override and Time Correction require reason and audit.
BR-05	Technician Check-in only assigned Visit; no overlapping active Work Session.
BR-06	Check-out requires Work Summary + Outcome + category-required CLEAN evidence; corrective resubmission always requires CLEAN evidence.

ID	Rule
BR-07	Only designated active Customer Acceptance Contact in same Site may Accept/Reject; Reject reason mandatory and creates Corrective Action.
BR-08	Accept -> COMPLETED; Supervisor closes only after reviewed Work Summary + Cost Summary -> CLOSED.
BR-09	Cancel Request before Convert; Visit before Check-in; WO before Customer Accept. Reason + audit required; terminal state rules apply.
BR-10	MISSED does not auto-create follow-up; Coordinator decides per D-15.
BR-11	Time Correction supports CHECK_IN, PAUSE_START, RESUME, CHECK_OUT; Technician/Team Lead request; Supervisor decides.
BR-12	Resolution SLA = 24h 24/7 from SUBMITTED; stops only Request REJECTED/CANCELLED or WO CLOSED/CANCELLED; prior breach preserved.
BR-13	Time Correction changes work-duration facts only; SLA Override is separate, authorized and audited.
BR-14	Duplicate check = same Site + Equipment/Location + Category within 24h against active set in D-12; warning not hard block; continuation reason required.
BR-15	Customer Reject -> Corrective Action -> Team Lead plan -> Supervisor approval -> corrective work -> summary review -> same contact re-accepts.
BR-16	Requester/Customer/Site mapping follows D-11; tenant/actor authority derived from authenticated context.
BR-17	SLA Risk event occurs at effective_due_at - 4 hours; Override recalculates risk_at; breach history remains sticky.
BR-18	Notification recipients follow Notification Recipient Matrix; failed delivery retries independently of committed business state.
BR-19	MISSED original Visit never transitions to CANCELLED/SCHEDULED; follow-up decision records outcome and may create a new linked Visit.

7. Locked Decisions and Revision Clarifications

Provenance: D-11 to D-15 are explicit revision clarifications introduced to close the cross-document audit gaps. They become locked baseline decisions only when this revision is formally approved.

ID	Topic	Locked / Proposed Baseline Decision
D-11	Identity / Customer Mapping	Requester = authenticated user that owns the Draft (mapped to created_by). Customer/ Company security scope = tenant_id . Site = site_id . Client must not supply tenant/actor as authority.
D-12	Duplicate Active Set	BR-14 “active Request” = SUBMITTED, UNDER_REVIEW, APPROVED, and CONVERTED only while its linked Work Order is not CLOSED/CANCELLED. DRAFT, REJECTED and CANCELLED are excluded.
D-13	SLA Risk Threshold	MVP “Near Breach / SLA Risk” is reached at 4 hours before effective_due_at . When an authorized SLA Override changes effective_due_at, risk_at is recalculated; an existing breached_at is never cleared.
D-14	Notification Routing	Every baseline notification has deterministic recipient(s) defined by the Notification Recipient Matrix. Delivery failure is retryable and must not roll back the business transition.
D-15	MISSED Decision Semantics	Original MISSED Service Visit remains immutable. Reschedule/Follow-up/Reassign creates a new SCHEDULED visit referencing the MISSED visit. Cancel decision records “no follow-up”; no new visit is created and the Work Order continues unless separately cancelled.

8. State Transition Reference IDs

Group	IDs	Coverage
Repair Request	ST-RR-001..008	Create/Edit, Submit, Route, Approve, Reject, Return, Cancel, Convert
Work Order	ST-WO-001..011	Schedule through Close/Cancel + Corrective cycle
Service Visit	ST-SV-001..009	Create/schedule, Check-in/out, Reschedule, Reassign, Cancel, MISSED decision
Work Session	ST-WS-001..004	Check-in, Pause, Resume, Check-out
Corrective Action	ST-CA-001..003	Create, Submit Plan, Approve Plan
Time Correction	ST-TC-001..003	Request, Approve, Reject
SLA Events	EV-SLA-001..005	Start, Risk, Breach, Override, Stop

9. NFR / Security / Constraints

Area	Testable Baseline
Security	Least privilege; backend authorization for role/tenant/site/ownership/assignment; IDOR denial without leakage; protected files; no sensitive data in client logs.
Performance	Normal List/Search <= 3 seconds at baseline load of 100 concurrent users and 100,000 records per Site using server-side filter/sort/paging and indexed query paths.
Reliability	State-changing commands are atomic, idempotent/retry-safe where applicable, and use optimistic concurrency; no partial transition.
File	PDF/JPG/JPEG/PNG <=10 MB/file; private storage; SHA-256; malware scan; only CLEAN file can satisfy mandatory evidence.
Usability	Responsive web; current Chrome and Edge; technician/customer flows usable on mobile; clear loading/empty/error/concurrency states.
Maintainability	Layered/equivalent separation; DTO validation; automated tests for BR/state transitions; traceable correlation ID.
Production dependency	Availability, backup, RTO/RPO, authentication mechanism, CORS/rate limit/provider settings must be finalized in Architecture/Deployment before Production.

10. First-release Acceptance Criteria

- FR-01/02: valid Draft/Submit works; duplicate warning obeys D-12/BR-14 and continuation reason is audited.
- FR-03/04/05: approval, Return to DRAFT, conversion, Service Visit actions obey role/state/scope and do not create duplicates.
- FR-06: only assigned Technician can manage Work Session; Check-out requires complete summary/outcome/evidence.
- FR-07: only designated Requester/Site Contact may Accept/Reject; reject creates traceable corrective cycle and same contact re-accepts.
- FR-08: 24h SLA starts/stops correctly; risk at T-4h; breach is sticky; Override and Time Correction are separate audited actions.
- FR-09: search/timeline/report/export are tenant/site scoped and prevent sensitive leakage.
- NFR: performance/browser/responsive/reliability test cases pass before release sign-off.

11. Baseline / Approval / Change Control

Revision v1.5 is a **Baseline Candidate**. Formal approval requires Business Owner / Project Manager name/date/ signature. After approval, any change to FR/BR/State/Data/Test or D-01..D-15 requires Change Request with impact on scope, timeline, cost, risk, Design, Development, Test and UAT.

12. Data Mapping & Validation

Business term	Canonical mapping / validation
Requester	Authenticated Draft owner; persisted as created_by. submitted_by snapshots submit actor.
Customer / Company	Security/business scope represented by tenant_id. tenant_id is server-derived, never trusted from client.
Site	site_id; active and within actor scope.
Equipment / Location / Category / Priority	Active master; category decides conditional Equipment/Location and evidence policy.
Customer Acceptance Contact	Exactly one per Work Order before AWAITING_CUSTOMER_ACCEPTANCE; source role only REQUESTER or SITE_CONTACT; active and same Site.
Cost Summary	Required and Supervisor-reviewed before CLOSED.
Audit	Append-only entity/id/action/actor/time/from/to/reason/old/new/correlation as applicable.

13. Action-level Permission

Action	Authorized role / Guard
Create/Edit/Submit Request	Requester; own DRAFT; valid fields/master/scope; duplicate rule.
Approve/Reject/Return	Assigned/authorized Approver; Reject/Return reason.
Create WO / Schedule / Assign	Coordinator; Request APPROVED; active team/site; current version.
Reassign/Reschedule/MISSED decision	Coordinator; reason; D-15 for MISSED.
Check-in/Pause/Resume/Check-out	Assigned Technician; valid Visit/Session state; no overlap.
Submit Work/Cost Summary	Team Lead; required facts/evidence complete.
Accept/Reject Work	Designated active Customer Acceptance Contact only; WO awaiting acceptance.
Corrective Plan / Close / Cancel WO	Supervisor per current state/guard; reason where required.
Time Correction	Technician/Team Lead request; Supervisor approves/rejects.
SLA Override	Supervisor with explicit permission; reason required; preserve original due/breach.

14. Master Traceability Matrix

FR	BR	State/Event	Use Case	Data	Test	API	UI
FR-01	BR-01,02,16	ST-RR-001	UC-RR-001	RR-001..020; ATT-*	TC-RR-001/002	RR-API-001/002; FILE-API-*	UI-010/011/012
FR-02	BR-02,14,16	ST-RR-002	UC-RR-002	RR-003..020; APR-*	TC-RR-003/004	RR-API-005	UI-011/012

FR	BR	State/Event	Use Case	Data	Test	API	UI
FR-03	BR-02,04	ST-RR-003..006	UC-RR-003/004	APR-*; AUD-*; NTF-*	TC-RR-005..009	RR-API-006..009	UI-020/012
FR-04	BR-03	ST-RR-008; ST-WO-000	UC-WO-001	WO-*	TC-RR-010/011	RR-API-010	UI-012/031
FR-05	BR-04,09,10,19	ST-SV-001..009	UC-WO-003/005/006/007/008/009	SV-*	TC-WO-002..005	WO-API-002..007	UI-032/033
FR-06	BR-05,06,11,13	ST-WS-001..004; ST-TC-*	UC-WO-012/016; UC-TIME-001	WS-*; TC-*; WSM-*; EVD-*	TC-WO-006/007; TC-TIME-001/002	WS-API-*; TIME-API-*	UI-040/041/080
FR-07	BR-07,08,15	ST-WO-004..010; ST-CA-*	UC-WO-020..024	CAC-*; ACC-*; CA-*; CST-*	TC-WO-008..012; TC-CST-001	ACC-API-*; CA-API-*; CST-API-*; WO-API-010/011	UI-050/060/070
FR-08	BR-12,13,17,18	EV-SLA-001..005; ST-TC-*	UC-SLA-001/002; UC-TIME-001	SLA-*; OVR-*; NTF-*	TC-SLA-001/002; TC-NTF-001/002	SLA-API-*	UI-090
FR-09	BR-02,16,18	AUD-CTRL-001	UC-WO-002; UC-REP-001	AUD-*; NTF-*	TC-WO-001; TC-REP-001; TC-SEC-001/002	REP-API-*; AUD-API-001	UI-001/010/030/100

15. Notification and MVP Reporting

Event	Trigger	Recipient(s)	Subject	Rule
NTF-SUBMITTED	Request Submitted	Assigned Approver(s)	Repair Request	Create in-app event after commit; routing failure remains SUBMITTED
NTF-RETURNED	Return for Correction	Requester	Repair Request	Reason shown; Request is DRAFT
NTF-APPROVED	Request Approved	Coordinator(s) in Site scope	Repair Request	Action required: Convert
NTF-REJECTED	Request Rejected	Requester	Repair Request	Terminal; SLA stopped
NTF-ASSIGN	Assign/Reassign Visit	Assigned Technician + Team Lead	Service Visit	Include schedule; no sensitive leak
NTF-MISSED	Visit Missed	Coordinator + Team Lead	Service Visit	Decision required
NTF-SUMMARY	Work Summary Submitted	Supervisor	Work Order	Review required
NTF-ACCEPT	Customer Accepted	Supervisor + Coordinator	Work Order	Close guard remains Cost/Review
NTF-REJECT-WORK	Customer Rejected	Team Lead + Supervisor + Coordinator	Work Order/Corrective Action	Corrective plan required
NTF-SLA-RISK	SLA Risk (T-4h)	Coordinator + Supervisor	SLA Record	No state change
NTF-SLA-BREACH	SLA Breach	Coordinator + Supervisor	SLA Record	breached_at set once
NTF-TIME	Time Correction Decision	Correction requester	Time Correction	Show approve/reject result
NTF-CANCEL	Cancel	Requester + current operational owner as applicable	Request/Visit/WO	Terminal/propagation effect summarized

MVP reports: work by status; near-breach/breached; Site/team; Cancel/Reject history. Filters: date range, Site, Category, Priority, Status; export must enforce the same tenant/site/PII rules.

16. UAT / Definition of Done

Feature is Done when linked functional/security/concurrency/notification/NFR tests pass with evidence. UAT sign-off requires all UAT=Yes cases passed, Critical/High defect = 0, formal trace evidence, and Business Owner/Test Lead sign-off.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

17. Performance NFR Clarification and Decision Boundary

The existing business NFR remains unchanged: **Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site**, with server-side filtering/sorting/paging and query paths designed for the stated access patterns.

PERF-DEC-01 - Performance Measurement Statistic (Pending Approval).
Business Baseline ระบุ “Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site” แต่ไม่ได้ระบุว่า 3 seconds หมายถึง Average, Maximum, p95 หรือ p99. จึง **ห้ามถือ p95 เป็น Requirement ที่อนุมัติแล้ว**. Recommendation สำหรับการอนุมัติคือ p95 end-to-end API response time <= 3.0 seconds; ก่อนอนุมัติ ให้บันทึก Average/p50/p95/p99/throughput/error rate ครบและรายงานผลเป็น diagnostic evidence โดยยังไม่ประกาศ Pass/Fail ของ release gate จาก percentile ใด percentile หนึ่ง.

No additional hard latency budget is created for Detail, Create/Update, State Transition, Dashboard, Report/Export, Timeline, Notification Inbox, File I/O or SLA Worker. Until separately approved, those paths are measured diagnostically and must still preserve correctness, security and resource safety.

18. Performance Traceability Extension

Area	Requirement / Business Source	Technical Design	Verification / Evidence
Normal List/Search	Existing NFR <=3s @ 100 concurrent / 100k records per Site	Architecture + DB/API/UI server-side bounded query path	TC-PERF-001/002; Average/p50/p95/p99 + throughput + DB/query/trace evidence; release statistic pending PERF-DEC-01
Security under load	Existing Security NFR / FR-09	Server-derived tenant/site scope before paging; no client authority	TC-SEC-001 + load companion evidence; no scope leakage
Report/Export	FR-09 functional scope only	API/UI/Architecture technical contract; exact size/format/async threshold not a locked requirement	TC-PERF-003 diagnostic + rows/bytes/time/memory
Worker / Notification	BR-17/18 semantics	Separate idempotent Worker + outbox; configuration tuned from evidence	TC-PERF-004 diagnostic + lag/backlog/retry/duplicate evidence
Files	<=10MB/private/CLEAN evidence baseline	Private object storage; authorization + scan	TC-PERF-005 diagnostic + bytes/duration/memory/scan backlog

Area	Requirement / Business Source	Technical Design	Verification / Evidence
Concurrency	Reliability NFR + rowversion semantics	ETag/If-Match + atomic transaction	TC-SEC-002 / TC-PERF-006 + conflicts/waits/no lost update

PTP-01 - Repeatable Load Test Profile (Technical Test Design)

Item	Technical Test Design - not a new Business Requirement
Baseline concurrency	100 concurrent virtual users - existing baseline.
Dataset	100,000 Repair Request records per Site - existing baseline. Related WO/Visit/Audit/Notification data must be representative enough to exercise joins and security scope.
Security scope	Use at least 2 tenants and 2 Sites in the test fixture where practical; every request is authenticated. Cross-scope negative traffic is measured separately.
Ramp / duration	Default proposal: 5-minute ramp-up, 30-minute steady-state, 5-minute ramp-down, plus one warm-up run. This may be adjusted in a versioned Test Plan and is not a business requirement.
Filters / sorting / paging	Use representative filters, common sort, first/middle/deeper pages and the same server-side tenant/site scope as production code.
Evidence	Build/commit, environment, DB migration, dataset seed/version, script version, Average/p50/p95/p99, throughput, unexpected error count, API/DB latency, CPU/memory, DB waits/blocking/slow-query evidence and trace samples.